

29

Integrated landfill mining

Challenge

Urban India generates about 1.5 lakh MT of waste daily out of which less than 10% is processed and all the rest is unscientifically dumped. There are about 4,500 dumpsites in urban India alone. The Solid Waste Management Rules of 2016 mandate reclamation of all these dumpsites with technologies which leave no residues. The country has been grappling with the challenge of proper management of municipal solid waste landfills from the days plastic was introduced in households. The problem is multifold (1) The waste is unsegregated and the Municipal bodies do not have infrastructure to segregate them (2) the waste is forcibly dumped in open dumpsites as a result (3) the waste ends up in landfills, giving no scope to the Municipal bodies to build new infrastructure, and also contaminating soil and underground water, polluting air, etc. (4) the neighbors protest when the Municipal bodies try to acquire new land for landfills. As India heads to become the world's largest populated country, the problem of haphazard dumping of municipal solid waste is increasing, causing damage to public health and environment.

Solution

Integrated landfill mining, a pro-poor innovation, ensures that the haphazardly dumped Municipal solid waste in open landfills is mined and completely removed, addressing the problems of air pollution, water pollution and soil contamination and helping provide a clean environment to the poor people living nearby these landfills. It provides a comprehensive solution to the Municipal bodies to outsource their dumpsites to Zigma Global Environ Solutions Private Limited and take back the reclaimed land upon completion of the project.

Key features:

1. Provides an integrated model wherein Municipal bodies need not invest in plant and machineries or any capital expenditure.

The company, Zigma Global Environ Solutions Private Limited will be taking care of all these things.2. The machines used are completely mobile, and can be completely dismantled and removed on completion of the project.3. The model is scalable depending on the quantity of waste present in the dumpsite.4. Baseline study of the implementation site is conducted to understand the underground water contamination, leachate, surface water contamination and soil contamination. Once the project is completed, a baseline study is again done to see the difference. The study reveals drastic (and continuing) reduction in the contamination levels. There are no emissions and zero residues on completion of the project. The company earns majorly from the processing charges collected from the ULBs. The company also earns revenue from the sales of aggregates like RDF, soil, stones, soil enricher, metal scrap, and glass scrap.

Implementation & Impact

The company has implemented various (completed and on-going) large-scale bio-mining projects such as revamping and clearing of existing dumped municipal solid waste in Thepperamanallur compost yard using bio-mining technology under Kumbakonam municipality, removal of legacy waste dumped on the banks of Sembakkam lake through bio-mining process under Sembakkam municipality, removal of legacy waste dumped on private land through bio-mining process on Design, Build, Own, and Operate (DBOO) concept in Pammal municipality, remediation of existing MSW dumpsite of Ajithsingh Nagar through bio-mining process in Vijayawada Municipal Corporation on Design, Build, Operate (DBO) concept, remediation of legacy waste located at Atladara utilizing various methodologies/technologies confirming to SWM Rules of 2016, and guidelines of Swachh Bharat Mission, Mahatma Gandhi Swachhata Mission and Hon. National Green Tribunal under Vadodara Municipal Corporation, dumpsite remediation of municipal solid waste at Noida sec 145 under NOIDA Authority, remediation/bio-mining of existing landfill/stabilised windrows located at Bhandewadi in Nagpur Municipal Corporation, removal of legacy waste dumped at Periya Eri through bio-mining in Pallavapuram municipality, removal of legacy waste dumped at Parivakkam dumpyard through bio-mining process and removal of additional legacy waste dumped on the banks of Parivakkam burial ground through bio-mining process on Design, Build, Own and Operate (DBOO) concept under Poonamallee municipality, revamping of dumpsite through bio-mining process on Design, Build, Own and Operate (DBOO) concept in Chidambaram municipality, removal of legacy waste dumped at Kanndapalyam dumpyard through bio-mining under Tambaram Municipality, remediation of existing MSW dumpsite at Ramapuram through bio-mining process under Tirupati Smart City Corporation Limited, reclamation of existing dump yard at Ariyamangalam through bio-mining process under Tiruchirapalli Smart City Ltd, revamping of existing dumped

garbage at Vendipalayam and Vairapalayam dumpyard through bio-mining process on DBFOO concept under Erode City Municipal Corporation, removing of dumpsite through bio-mining to recover the land at Devakottai road compost yard in Karaikudi Municipality, removing the legacy waste dumped in the compost yard through bio-mining process at Karur Municipality, revamping of dumpsite through bio-mining to recover the land at Kamiyanpettai and Panchayankuppam compost yard in Cuddalore Municipality and removal of legacy waste 200000 cu.m through bio-mining process to reclaim the existing site at Murugabavanam in Dindigul City Municipal Corporation.

Kumbakonam Dumpyard Reclamation Project

Before



After



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